Futures Expiration Day Effects on Spot Market Volatility: Evidence from the NSE

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Abstract

The present study seeks to investigate the futures contracts expiration day effects on price reversal and volatility in Nifty as well as individual stocks by employing GARCH framework. The results confirm presence of upward pressure in volatility of Nifty at both mean and variance level, at time of futures contracts expiration day. However there is neither a positive nor negative price (and volatility) reversal following the expiration of derivative contracts. In case of individual stocks, volatility has been increased at time of expiration day of futures contracts in most of the cases. The results however could not find price reversal for most of the individual stocks.

Keywords: GARCH Framework, Expiration Day Volatility Effect, Price Reversal Effect

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I. Introduction

Index derivatives are frequently accused of creating undesirable effects on the market for the underlying securities. The expiration day effect is one of these effects. It is a temporary distortion of the underlying stock prices due to an order imbalance of trades on those days when the relevant index futures contracts expire. This abnormal price movement or distortion of the underlying stock prices of index stocks of the expiration days is undesirable. Traders who appear with market orders to buy (sell) when an expiration is pushing prices up (down) may be hurt. Though the occurrence of such event can be predictable, the direction of the price movement and the magnitude of order imbalances on expiration days remain uncertain. Thus, studies on the expiration-day effect on the underlying individual stocks are needed. Such studies are expected to provide benefit to the investors on the spot market. Investors, especially small investors, may be forced to stay away from the spot market temporarily if they do not have a good understanding of the expiration-day effect. Therefore, it is important to examine the effect of futures expiration day on spot market volatility.

II. Futures Expiration Day Volatility Effect

The combined effect of factors like existence of index arbitrage opportunities, cash settlement of index futures and the stock market procedures may cause expiration-day volatility effects. At the expiration of the futures contract, the index arbitrage positions are frequently unwound. The futures self-liquidate through cash settlement at the closing index level as and when index futures expire at the close. On the other hand, through trades in the marketplace, the stock position must be liquidated. An arbitrageur is long the underlying stocks

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and short the index futures contract must sell the underlying stocks at their closing prices. As long as the stocks sold at the same prices used in calculating the index value cash settling the futures contract, the arbitrageur exits his position recklessly, independent of the level of the prices at which the stocks are sold. Volatility effects are possible, if many arbitrageurs liquidate positions at the same time and in the same direction.

Stock price volatility effects of expiration day depend in part on the stock market procedures with accommodating order imbalances that may arise when arbitrage positions are unwound. If the underlying market for the index stocks is deep and if suppliers of liquidity are quick respond to selling or buying pressure, the price effects of large arbitrage unwinding are small. If unjustified price effects are known to occur, knowledgeable investors would stand ready to buy under priced stocks and sell overpriced stocks-actions that would normally limit price effects fall within the bounds of transaction costs. If market mechanisms are not well designed to offset sudden imbalances, however, the volatility effects may be substantial. In the case of index futures contracts that settle at the close, arbitrage positions must be unwound at closing prices.

III. The Futures Expiration Day Price Reversal Effect

Due to price pressure when index derivatives expire, the arbitrageurs liquidate their positions or else the speculators attempt to raise or lower the prices in the spot markets in order to increase their profits. In both the cases the prices in the spot markets will deviate from their equilibrium levels. This phenomenon, however, is only temporary as price pressure will be released suddenly when the expiration days pass and stock market prices will recover in the direction of the equilibrium price. That is, if the stock index goes up on the expiration day, then it will go down after the expiration day and the price pressure will disappear. If the stock index goes down on the expiration day, then it will go up after the expiration day.

The remaining part of study is organized as follows: Section IV presents review of empirical literature on futures expiration day effects. Section V deals with the

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main characteristics of NSE 50 and data set used. Section VI presents methodology Section VII discuss the empirical results and the final section gives the concluding remarks

IV. The Expiration Day Effect on Spot Market Volatility: Some Empirical Issues.

There are some studies which have investigated into the effects of derivative contracts of expiration day on price reversal and volatility of underlying markets. However, these studies remain inconclusive. For example Stoll and Whaley (1987), Day and Lewis (1988), Chamberlain *et al.* (1989), Pope and Yadav (1992), Stoll and Whaley (1997), and Diz and Finucane (1998), found presence of higher spot market volatility at time of futures contracts expiration day. In contrast to the above studies, Kan (2001) had not observed a significant price volatility and price reversal in Hong Kong market. For the same market, Chow, Yung and Zhang (2003) observed a negative price effect and some return volatility of cash prices on account of expiration day effects, while Chan and Williams (1994) have found no effect of expiration on mean retunes and volatility of underlying asset prices in the cash market. Therefore, the nature of the impact of expiration of derivatives on the underlying cash prices remains an open question.

The expiration effects of TAIFEX index derivatives on the underlying stock market were examined by Chen and Chen (2002). They used daily data between 1998 and 2002. They conducted pooled F-test for an independent sample to compare the standard deviations of the returns in order to test whether the stock market is more volatile on the expiration day than on the non-expiration day. Their empirical findings showed that there were no significant volatility effects during expiration day, but demonstrated that expiration effects got stronger as more relative index derivatives were listed on the TAIFEX. Meanwhile, the expiration effects seemed to shift to the opening of the settlement day. In general, the expiration effects in Taiwan are not as significant as those in the U.S. markets, but are stronger than those in Hong Kong market. The special

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settlement procedures adopted by the TAIFEX might be the reason for the difference.

The effect of expiration day of the Index futures and on the trading volume, variance and price of the underlying shares was investigated by Keran (2003). He used both daily and high frequency (5 minutes and 10 minutes) data on S&P CNX Nifty Index from January 2003 through September 2006. Employing GARCH framework to capture time varying volatility in data, the study had shown that while there was no pressure – downward or upward - on index returns, the volatility was indeed significantly affected by the expiration of contracts for intra- day data. This effect, however, was not demonstrated in daily data.

The impact of expiration of Hang Sang Index (HSI) derivatives on the underlying cash market in Hong Kong was investigated by Ying-Foon Chow et al. (2003). Employing Comparison - Period approach, they found that expiration days in Hong Kong might be associated with a negative price effect and some return volatility on the underlying stock market, but there was no evidence for price reversal after expiration.

The effect of expirations of Hang Seng index derivatives on stock market volatility was investigated by Bollen and Whaley (1999). They conducted variance test and computed growth rates in order find out the volatility effect. They concluded that there was higher volatility at time of expiration day of futures contracts.

The issue of how the remaining time to the expiration date of derivative markets affects the volatility of the IBEX 35 index (expiration effect) and its futures (maturity effect) was examined by Arago and Fernandez (2002). They used both daily closing price of IBEX 35 index and settlement price of futures contracts. They employed bivariate ECM-GARCH model including dummy variables that expresses the time left to expiration. The results of GARCH model showed that, during the week of the expiration day, the conditional variance of both markets increased without presenting any significant behavior in the correlation level.

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During the second week of expiration day, however, the conditional variance diminished and the degree of correlation between the two markets increased.

Hancock (1993) used minute-by-minute S&P 500 index values for the period from 30 April 1987 to 24 July 1989 to investigate the triple witching hour effect. Employing variance test, he found expiration-day effect in the underlying spot market. He further indicated that the remarkable increase or decrease in the underlying spot market volatility around the triple witching hour in the USA could be attributed to investors' responses to new information, especially important economic news which is often disclosed on Fridays, that is, the same day as the expiration of the futures contracts.

The impact of futures expiration day on return volatility and price reversal of Toronto Stock Exchange (TSE) was investigated by investigated Chamberlain *et al.* (1989). They used high frequency data from November 1985 to May 1987. They observed that the rate of return during the last half- hour of trading was significantly more volatile on expiration days than that on other non-expiration Fridays. Besides that, the expiration returns also tended to be reversed during the first half-hour of trading on the following Monday.

The issue of whether changes in expiration settlement procedures reduce spot market volatility has been by investigated Herbst and Maberly (1990). They also tested whether there is any change in triple witching hour volatility and expected price reversal (EPR), in two different studies respectively, before and after the new settlement procedure. Their results are consistent with the findings of Stoll and Whaley (1991). They showed that volatility on the expiration days was significantly reduced, but the volatility was shifted within a day together with a significant increase in the first hour volatility.

Bessembinder and Seguin (1992), using daily data on the S&P 500 index from January 1978 to September 1989, investigated the expiration-day volatility effect and reached different conclusions. Employing a regression of daily S&P 500 return standard deviation with the independent variables such as spot-trading

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volume and futures trading dummy and days to expiration dummy, they reported that the equity volatility was slightly higher on the expiration days.

In a similar study, Schlag (1996), found a significant increase in volatility. While analyzing the impact of options expiration, the study concluded an increase in trading activity as well as in volatility for the ten-minute settlement period of index options. However, volatility remained unchanged around the expiration of a futures contract.

Empirical studies of the futures expiration day effect on volatility have mixed results. Further, most of the previous studies on expiration day effects have focused exclusively on the well-developed stock markets, usually the US markets. In addition, not even a single study has been conducted to investigate the futures expiration day effects on individual stocks, particularly in perception of India's single stocks futures. Therefore, this study makes an attempt to analyze the effect of expiration stock futures on volatility and price reversal of underlying stocks in the Indian Stock Market.

V. Data Description

NSE is a leading stock exchange of India with full transparency and regulatory framework. The CNX Nifty comprises of 50 most liquid individual stocks and it represents 22 sector of Indian economy. NSE has introduced index futures contracts trading on June 12, 2000. S&P CNX Nifty futures contracts expire on the last Thursday of the expiry month. If the last Thursday is a trading holiday, the contracts expire on the previous trading day. The study has selected twenty one individual stocks out of fifty, to examine expiration day effect on volatility of individual stocks. The study could not cover all the 50 individual stocks, because some of them have introduced stocks futures only recently.

To make the study representative of the whole market, a leading private bank namely, the ICICI Bank has been included in the sample. The derivatives contracts of ICICI bank were introduced on January 31, 2003. The names of the

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21 stocks along with their introduction dates of the options and futures are given in table 1

Secondary data has been used in this study. The averages of daily closing prices of 15 individual stocks from November 9, 2001 to February 28, 2008, are collected to examine the impact of derivatives expiration day on volatility of individual stocks. Daily closing prices of remaining six stocks are collected from January 31, 2003 to February 28, 2008. The study uses daily closing price of Nifty from July 12, 2000 to February 28, 2008 to investigate futures expiration day effect on volatility of nifty.

VI. Methodology

The presents study investigates expiration day effect on spot price volatility as well as individual stocks volatility using GARCH model. As noted earlier, the previous studies on expiration day effects contend themselves with comparing unconditional means and variances by applying classical hypothesis testing procedures like t- and F tests. However, they do not capture for stylized features of financial data such as time varying volatility, skewness and excess kurtosis. The GARCH model proposed by Bollerslev (1986) has been widely used in finance literature to explicitly capture these empirical features. The GARCH framework not only aims to capturing the time varying nature of volatility but also able to link information arrival and volatility. The standard GARCH (p,q) model is given by:

$$R_t = a_0 + a_1 R_{t-1} + \varepsilon_t \qquad \ldots (1)$$

$$\varepsilon_t \mid \Psi_{t-1} \sim N(0, h_t)$$

$$h_t = \beta_0 + \sum_{i=1}^{p} \beta_i \varepsilon_{t-i}^2 + \sum_{j=1}^{q} \gamma_j h_{t-j} \qquad \ldots (2)$$

where R_t is the logarithmic daily return, ε_t is the error in the conditional mean equation and Ψ_{t-1} is the set of information available at time t-1. The coefficients of the moving average component of conditional variance, β_i, are typically interpreted as news coefficients that measure the impact of recent news on volatility; γ_j, the coefficients of the autoregressive component of conditional

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variance, are similarly interpreted as persistence coefficients that measure the impact of less recent or old news on current volatility.

Since study proposed to examine the effect of futures expiration day on the conditional mean and variance, the study include a dummy variable for the expiration day in GARCH model. Further, study also intends to confirm the presence of price reversal, study also include another dummy to represent the day after the expiration day. For each of the return series, the conditional mean equation and variance are modified as:

$$R_t = a_0 + a_1 R_{t-1} + \Phi_1 ExpD_t + \Phi_2 NextD_t + \varepsilon_t \qquad \ldots (3)$$

$$\varepsilon_t \mid \Psi_{t-1} \sim N(0, h_t)$$

$$h_t = \beta_0 + \beta_1 \varepsilon^2_{t-1} + \gamma_1 h_{t-1} + \delta_1 ExpD_t + \delta_2 NextD_t \qquad \ldots (4)$$

where $ExpD_t$ is the dummy that takes the value 1 for expiration days and zero other wise, while $NextD_t$ is the dummy variable that takes the value 1 for all days that immediately follow expiration and zero otherwise. If the coefficient of dummy $ExpD_t$ turns up significant and negative (positive), then one can affirm the presence of downward (upward) pressure on the conditional return during the expiration period. Further, if coefficients of both $ExpD_t$ and $NextD_t$ turn up significant and have different (same) sign, presence of price reversal (price continuation) can be validated.

The standard ARMA-GARCH model with as specified in equations (1) to (4) with normality assumption, sufficiently captures time-varying volatility; it might however be ineffective for capturing the excess kurtosis that might be present in the raw returns. If the conditional mean and conditional variance are correctly specified, the quasi-maximum likelihood estimator is consistent under the assumption that errors follow normal distribution. Though these estimators are unbiased, they are inefficient; the degree of inefficiency increases with the degree of departure from normality (Engle et al 1991). Further, it may be expected that excess kurtosis displayed by the residuals of a standard GARCH model will be reduced when a more appropriate distribution is used. Student-'t' (Bollerslev

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1987, Beine et al 2000) and Generalized Error Distribution (Nelson 1991, Kaizer, 1996) are two fat-tailed distributions that are widely used in empirical literature in place of normal distribution. In this study, we use Student-'t' distribution wherever normal distribution gives suspicious results.

VII. Empirical Analysis

The basic descriptive statistics of the Nifty are given in table 2. The table shows that the standard deviation of Nifty returns of expiration days is higher than non-expiration days. It indicates higher volatility at time of expiration day of futures contracts. Nifty return series are skewed and highly leptokurtic compared to normal distributions as measured by skewness and kurtosis.

Table 2 also provides preliminary evidence in favour of time-varying volatility for Nifty return series. In order to test the null hypothesis of independence, the portmanteau test of Ljung-Box (1978) is used. The LB- Q statistics show that high and significant autocorrelation in returns and squared returns up to lag 16, indicating the presence of strong conditional heteroscedasticity in the series. Further, LM tests observed the presence of ARCH effects in Nifty series. Before proceeding further, it may be necessary to verify that the Nifty returns and individual stock returns are stationary or not, using Augmented Dickey-Fuller and PP tests. Table 3 provides unit root results suggesting that the null hypothesis of unit root is strongly rejected in all cases.

Table 4 provides the descriptive statistics for all the individual stocks for both expiration and non expiration days. The daily mean returns for fifteen individual stocks are positive except in case of CIPLA, DRREEDY, INFOSYS, ITC, RANBAXY and WIPRO particularly during expiration days. In case of non expiration days, daily mean returns for seventeen individual stocks are also positive with exception of ACC, CIPLA, HEROHONDA and NATIONALU. The standard deviation of 17 individual stocks marginally increased in expiration days as compared to non expiration days. However, value of standard deviation for stocks such as CIPLA, HDFC, M&M and ONGC are same during both expiration and non expiration days.

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Table 4 also shows that the value of kurtosis exceed more than 3 for most of the individual stocks both on expiration and non expiration days. This indicates that returns of most individual stocks deviate from normal distribution. The values of skewness are negative in most of individual stocks implying that the frequency distribution of returns series is negatively skewed or having longer tails to the left, both during expiration days as well as non expiration days.

Table 5 presents result of GRACH analysis. The results reveal that coefficients of expiration day dummies in the mean and variance equation are positive and significant indicating there has been increase in volatility of Nifty at both mean and variance levels. However, the coefficients of $NextD_t$ have turned out to be insignificant. Hence, there is neither a positive nor negative price (and volatility) reversal following the expiration of derivative contracts. In other words, there is no reversal in either the mean or volatility of returns immediately after the expiration day.

The result of student't' distribution of GARCH model are reported in table 6. The results show that coefficient of expiration day dummy in the mean equation is positive and insignificant (0.0027). It implies that there is no higher volatility at mean level. The same is positive and significant (0.819) in the variance equation, indicating higher volatility at variance level at the time of futures contracts expiration day. However, the price reversals are not found for Nifty index. This is shown by the coefficient of $NextD_t$ which is positive and insignificant.

The study also investigates the effect of futures expiration day on volatility of individual stocks. The empirical results are given in tables 7 and 8. The results show that there has been an increase in volatility of seventeen stocks at the time of stock futures contracts expiration day. This is shown by coefficients of expiration day dummies which are positive and significant. However, there is no significant futures expiration day volatility effect, particularly, for stocks such as CIPLA, HDFC, M&M and ONGC.

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The tables also show that the coefficients of $NextD_t$ for sixteen individual stocks are found to be insignificant implying that there is no price reversal for these stocks either at mean or variance levels. However, price reversal is found for stocks such as BPCL, INFOSYS, ITC, SATYAM and WIPRO. This is because of coefficients of NextD are significant in both mean and variance equation. Similar results are observed in student't' distribution of GARCH model (table 8).

VIII. Conclusion

The study investigates the effect of futures contracts expiration day on price reversal and volatility in Nifty as well as in individual stocks. The study employed both normal and student t' distribution of GARCH models to capture time varying volatility and excess kurtosis in the data. The results showed that there is significant expiration day effect in National Stock Exchange of India. Further, the results reveal that there has been higher stock market volatility at time of futures contracts expiration day. This is due to the fact that index arbitrageurs unwind arbitrage positions in the stock market. While the cash settlement feature used for index derivatives allows derivative positions to be self-liquidated, stock positions must be liquidated through trades in the marketplace. If many arbitrageurs liquidate at the same time and in the same direction, price volatility is possible. However, the results show that there is no stock price reversal in either the mean or volatility of returns immediately after the expiration day. In case of individual stocks, volatility of seventeen stocks has been increased at the time of expiration day of futures contracts except in case of CIPLA, HDFC, M&M and ONGC. Further, the study could not find price reversal for most of the individual stocks.

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Table 1: Sample Stocks and Introduction Date of Options and Futures

Name of the Stock		Date of introduction of options	Date of introduction of futures
ACC	ACC	2-July-01	9-Nov-01
BHEL	BHEL	2-July-01	9-Nov-01
BPCL	BPCL	2-July-01	9-Nov-01
CIPLA	CIPLA	2-July-01	9-Nov-01
Dr. Reddy's Laboratories Ltd.	DRREEDY	2-July-01	9-Nov-01
GRASIM	GRASIM	2-July-01	9-Nov-01
Housing Development Finance Corporation Ltd	HDFC	2-Jul-01	9-Nov-01
Infosys Technologies Ltd.	INFOSYS	2-July-01	9-Nov-01
I T C Ltd	ITC	2-July-01	9-Nov-01
Mahindra & Mahindra Ltd	M&M	2-July-01	9-Nov-01
Ranbaxy Laboratories Ltd	RANBAXY	2-July-01	9-Nov-01
Reliance Industries Ltd	RELIANCE	2-July-01	9-Nov-01
Satyam Computer Services Ltd	SATYAM	2-July-01	9-Nov-01
State Bank of India	SBI	2-July-01	9-Nov-01
Tata Power Co. Ltd	TATAPOWR	2-July-01	9-Nov-01
HCL Technologies Ltd	HCL	31-Jan-03	31-Jan-03
Hero Honda Motors Ltd	HEROHONDA	31-Jan-03	31-Jan-03
ICICI Bank Ltd.	ICICI	31-Jan-03	31-Jan-03
National Aluminium Co. Ltd.	NATIONALLU	31-Jan-03	31-Jan-03
Oil & Natural Gas Corporation Ltd.	ONGC	31-Jan-03	31-Jan-03
Wipro Ltd	WIPRO	31-Jan-03	31-Jan-03

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Table 2: Descriptive Statistics on Nifty Index

Parameter	Non expiration Days (1822)	Expiration Days (91)
Mean	0.00075	0.014807
Std. dev	0.01567	0.069205
Skewness	-0.707825	-0.530775
Kurtosis	8.492145	3.802723
Jarque bera	2442.06	42.37
Residual diagnostics	Test statistic	P. value
L B-Q(16)	64.790	0.0000
LB^2-Q(16)	719.914	0.0000
LM- (10)	435.096	0.0000
F- Test	56.0157	0.0000

Table 3: Unit Root Test Statistics

Name of the stock	ADF in level	ADF in First Differences	PP in level	PP in First Differences
ACC	-29.660* (0.0000)	-19.536* (0.0000)	-38.475* (0.0000)	-51.529* (0.0001)
BHEL	-18.521* (0.0000)	-18.373* (0.0000)	-27.476* (0.0000)	-87.986* (0.0001)
BPCL	-37.997* (0.0000)	-17.580* (0.0000)	-37.994* (0.0000)	-40.783* (0.0001)
CIPLA	-28.528* (0.0000)	-12.853* (0.0000)	-26.528* (0.0000)	-86.851* (0.0000)
DRREDDY	-39.522* (0.0000)	-18.211* (0.0000)	-39.131* (0.0000)	-53.939* (0.0001)
GRASIM	-12.890* (0.0000)	-20.690* (0.0000)	-13.124* (0.0000)	-92.679* (0.0001)
HDFC	-30.686* (0.0000)	-19.943* (0.0000)	-41.579* (0.0000)	-65.235* (0.0001)

Figures in parentheses are P values. * denotes rejection of null hypothesis at 1 % level of significance

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Table 3 (contd.,)

Name of the stock	ADF in level	ADF in First Differences	PP in level	PP in First Differences
INFOSYS	-28.708* (0.0000)	-16.753* (0.0000)	-28.715* (0.0000)	-50.988* (0.0001)
I T C Ltd	-39.787* (0.0000)	-15.915* (0.0000)	-39.385* (0.0000)	-79.318* (0.0000)
M& M Ltd.	-37.619* (0.0000)	-20.231* (0.0000)	-37.563* (0.0000)	-75.075* (0.0001)
RANBAXY	-38.789* (0.0000)	-21.232* (0.0000)	-38.764* (0.0000)	-52.248* (0.0001)
RELIANCE	-40.540* (0.0000)	-20.216* (0.0000)	-39.578* (0.0000)	-50.120* (0.0001)
SATYAM	-39.168* (0.0000)	-16.844* (0.0000)	-38.492* (0.0000)	-47.100* (0.0001)
SBI	-29.678* (0.0000)	-19.820* (0.0000)	-37.824* (0.0000)	-41.333* (0.0001)
TATAPOWER	-31.118* (0.0000)	-13.514* (0.0000)	-36.419* (0.0000)	-37.971* (0.0001)
HCL	-35.552* (0.0000)	-16.098* (0.0000)	-35.631* (0.0000)	-55.189* (0.0001)
HEROHONDA	-23.448* (0.0000)	-18.343* (0.0000)	-36.004* (0.0000)	-51.988* (0.0001)
ICICI	-15.568* (0.0000)	-9.452* (0.0000)	-14.998* (0.0000)	-28.564* (0.0001)
NATIOANLALU	-33.703* (0.0000)	-17.861* (0.0000)	-33.644* (0.0000)	-42.062* (0.0001)
ONGC Ltd.	-22.539* (0.0000)	-11.235* (0.0000)	-23.471* (0.0000)	-68.757* (0.0001)
WIPRO	-37.388* (0.0000)	-17.945* (0.0000)	-37.243* (0.0000)	-49.971* (0.0001)

Figures in parentheses are P values. * denotes rejection of null hypothesis at 1 % level of significance

Table 4: Descriptive Statistics on Individual Stocks

Name of	Return	Numbe	Mean	S.D	Skewnes	Kurtosis

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the Stock	*	r of Days			s	
ACC Ltd	Non ED	1510	-0.0011	0.0232	-0.465	7.619
	ED	76	0.02081	0.1014	-0.492	10.394
BHEL	Non ED	1510	0.00181	0.0304	-7.102	159.871
	ED	76	0.03695	0.1228	-1.183	10.394
BPCL	Non ED	1510	0.00066	0.0271	0.027	8.82
	ED	76	0.01216	0.1236	-0.506	3.685
CIPLA	Non ED	1510	-0.0011	0.0517	-22.897	652.223
	ED	76	-0.0068	0.0519	-5.131	34.301
DRREDDY	Non ED	1510	0.0004	0.0582	-10.326	258.22
	ED	76	-0.0068	0.0694	-1.898	10.389
GRASIM	Non ED	1510	0.03032	0.1051	0.169	3.933
	ED	76	0.00259	0.1652	-0.253	4.896
HDFC	Non ED	1510	0.00093	0.0283	-9.937	246.814
	ED	76	0.00084	0.0281	-2.918	20.429
INFOSYS	Non ED	1510	0.00045	0.0463	-19.435	537.4
	ED	76	-0.0118	0.1862	-4.598	29.187
I T C Ltd	Non ED	1510	0.00082	0.0702	-34.441	1285.79
	ED	76	-0.0159	0.3024	-7.633	63.642
M& M Ltd.	Non ED	1510	0.03138	0.0306	-7.559	173.306
	ED	76	0.02687	0.0307	-1.069	9.477
RANBAXY	Non ED	1510	0.0003	0.0195	-11.952	268.012
	ED	76	-0.0075	0.1508	-2.719	14.352
RELIANCE	Non ED	1510	0.00148	0.0219	-1.749	27.156
	ED	76	0.02897	0.0848	-0.451	3.917
SATYAM	Non ED	1510	0.00071	0.0331	-5.98	129.237
	ED	76	0.00892	0.0471	0.063	3.869

* ED – Expiration Days

Table 4 (contd.,)

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Name of the Stock	Return	Number of Days	Mean	S.D	Skewness	Kurtosis
SBI	Non ED	1510	0.00016	0.0224	-0.03616	6.961
	ED	76	0.03041	0.1095	-0.1516	-3.297
TATAPOWER	Non ED	1510	0.00161	0.0265	-0.376	12.901
	ED	76	0.03237	0.1236	-0..086	3.295
HCL	Non ED	1119	0.00042	0.0349	-7.604	158.892
	ED	61	0.00956	0.1505	-2.77	16.612
HEROHONDA	Non ED	1119	-0.0009	0.0233	0.565	5.816
	ED	61	0.07933	0.0793	0.241	3.429
ICICI	Non ED	1119	0.03471	0.0478	-0.093	56.12
	ED	61	0.04987	0.0658	-0.011	26.452
NATIOANLALU	Non ED	1119	-0.0013	0.0296	-0.118	7.587
	ED	61	0.02829	0.1244	-0.439	3.699
ONGC Ltd.	Non ED	1119	0.00083	0.0826	-2.822	40.857
	ED	61	0.01706	0.0826	-0.227	3.699
WIPRO	Non ED	1119	0.00096	0.0443	-14.21	314.565
	ED	61	-0.0196	0.1927	-3.587	18.558

* ED – Expiration Days

Table 5: Empirical Results of Futures Expiration Day Effect on Volatility of Nifty with GARCH (1, 1) Model: Normal Distribution

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Mean Equation			
Parameters	**Coefficients**	**Z-statistic**	**p-value**
a_0	0.000113	4.036*	0.0000
a_1	0.112789	4.332*	0.0000
Φ_1	0.022139	3.552*	0.0001
Φ_2	0.000202	0.716	0.2100
Variance Equation			
β_0	0.000167	7.064*	0.0000
β_1	0.199516	10.935*	0.0000
γ_1	0.728276	31.383*	0.0000
δ_1	0.983146	2.742*	0.0114
δ_2	0.000053	0.1226	0.4721
Residual diagnostics	**Test statistic**	**P. value**	
L B-Q(16)	9.162	0.742	
LB^2-Q(16)	12.824	0.819	
LM- (10)	8.799	0.951	
F- Test	0.879	0.652	

Note: * Indicates 1% significance level. The GARCH (1, I) model corresponds to equations (1) and (4)

Table 6: Futures Expiration Day Effect on Volatility of Nifty with GARCH (1, 1) Model: Student-t Distribution

Mean Equation			
Parameters	**Coefficients**	**Z-statistic**	**p-value**
a_0	0.001210	4.447*	0.0000
a_1	0.104533	4.260*	0.0000
Φ_1	0.002782	1.308	0.0212
Φ_2	-0.000081	-0.912	0.0854

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Variance Equation			
β_0	0.000318	4.201*	0.0000
β_1	0.173863	6.847*	0.0000
γ_1	0.567826	24.943*	0.0000
δ_1	0.819411	2.713*	0.0112
δ_2	0.000112	0.531	0.5896
Residual diagnostics	**Test statistic**	**P. value**	
L B-Q(16)	8.725	0.926	
LB^2-Q(16)	5.876	0.754	
LM- (10)	11.727	0.804	
F- Test	1.173	0.803	

Note: * Indicates 1% significance level. The GARCH (1, I) model corresponds to equations (1) and (4)

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Table 7: Empirical Results of Futures Expiration Day Volatility Effect with GARCH (1, 1) Model: Normal Distribution

	Mean Equation				Variance Equation				
Name of the stock	Constant	L.return	ExpD	NextD	Constant	ARCH(1)	GARCH(1)	ExpD	NextD
ACC LtD	0.0001	0.0036	0.0115*	0.0016	0.01836	0.132	0.833	0.7072*	0.0036
	(2.188)	(0.155)	(5.215)	(0.708)	(4.562)	(6.521)	(48.632)	(2.459)	0.856)
BHEL	0.0009	0.0915	-2.0111*	0.0038	0.00090	0.222	0.620	0.8057*	0.0034
	(3.615)	(3.176)	(-5.562)	(0.449)	(7.002)	(9.983)	(15.674)	(29.33)	(0.693)
BPCL	0.0003	0.0515	0.0082*	0.0008*	0.02185	0.060	0.905	0.0625*	0.5451*
	(0.509)	(1.969)	(3.013)	(2.576)	(3.825)	(10.087)	(87.753)	(9.732)	(10.288)
CIPLA	-0.0001	0.0425	0.0014	0.0002	0.00074	0.036	0.794	0.0003	0.0006
	(-0.395)	(1.302)	(0.412)	(0.213)	(29.479)	(2.632)	(79.112)	(0.119)	(0.465)
Dr. Reddy's	-0.0007	0.0173	0.8001*	0.0003	0.00024	0.141	0.846	0.8001*	0.0010
	(0.436)	(0.597)	(5.019)	(0.625)	(22.395)	(5.436)	(110.99)	(5.080)	(0.776)
GRASIM	0.0014	0.0589	-2.002*	-0.0892	5.25E-05	0.149	0.794	0.9082*	0.0001
	(3.046)	(2.138)	(-3.900)	(-1.845)	(5.382)	(10.770)	(46.495)	(2.115)	(0.116)
HDFC	0.0007	0.0320	-0.0056	-0.0010	0.00023	0.015	0.731	0.0007	-0.0836
	(0.771)	(1.128)	(-1.089)	(-0.498)	(3.932)	(4.267)	(9.276)	(0.061)	(-0.132)
INFOSYS	-0.0026	0.0081	0.0087*	-0.0071*	0.00212	0.263	0.597	0.7623*	-0.8015*
	(-0.427)	(0.121)	(2.128)	(-2.889)	(19.314)	(6.423)	(20.009)	(5.112)	(-7.623)
ITC	0.0044	0.0022	0.9002*	-0.0092*	6.45E-05	0.158	0.682	2.5613*	-4.2142*
	(0.330)	(0.934)	(7.029)	(-5.692)	(7.098)	(4.001)	(15.092)	(12.51)	(-9.267)
M&M	0.0009	0.0618	0.0002	0.0001	0.00070	0.475	0.278	-0.0032	-0.0354
	(0.658)	(2.290)	(0.136)	(0.332)	(2.532)	(6.262)	(7.461)	(-0.957)	(-1.33)

Note: ExpD and NextD indicate volatility effect and price reversal effect. Figures in parentheses are t values. * indicates 1 % level of significance.

Table 7 (contd.,)

Mean Equation	Variance Equation

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Name of stock	Constant	L.return	ExpD	NextD	Constant	ARCH(1)	GARCH(1)	ExpD	NextD
RANBAXY	-0.0063	0.0439	0.5001*	-0.0091	0.00652	0.164	0.300	0.0049*	-0.0003
	(-0.726)	(0.621)	(2.0283)	(-1.26)	(18.344)	(5.631)	(7.814	(9.499)	(-1.258)
RALIANCE	0.0013	0.0078	0.6009*	0.0919	0.00016	0.342	0.426	3.5267*	0.0007
	(3.005)	(0.249)	(8.018)	(1.426)	(8.790)	(23.456)	(12.120)	(2.897)	(0.374)
SATYAM	-0.0001	0.0063	0.0059*	0.0071*	0.00164	0.245	0.589	0.8018*	-0.0080*
	(-1.745)	(0.856)	(2.596)	(-2.897)	4.391)	(7.112)	(5.065)	(12.053)	(-4.179)
SBI	0.0015	0.0307	6.0027*	0.0006	2.56E05	0.087	0.887	0.9125*	0.0076
	(2.892)	(1.226)	(-4.249)	(0.568)	(3.137)	(8.614)	(64.125)	(4.211)	(0.612)
TATAPOWER	0.0001	0.1084	0.0465*	-0.0015	0.2374	0.155	0.804	2.113*	0.0006
	(2.694)	(4.108)	(3.369)	(1-.589)	(4.649)	(9.986)	(42.020)	(9.243)	0.442)
HCLTECH	0.0006	0.1254	0.0065*	0.0013	7.27E05	0.198	0.873	2.785*	-0.0011
	(0.757)	(3.426)	(2.041)	(0.168)	(1.831)	(8.492)	(65.180)	(8.421)	(-1.211)
HEROHONDA	0.0004	-0.02121	0.0078*	0.0001	0.5906	0.145	0.737	0.0034*	0.0002
	(0.742)	(-0.702)	(2.462)	(0.442)	(5.285)	(6.037)	(19.136)	(2.432)	(0.630)
NATIONALU	0.0021	0.0495	0.0024*	-0.0856	0.9856	0.110	0.858	0.1006*	0.0002
	(2.294)	(1.669)	(2.644)	(-0.452)	(8.521)	(9.770)	(64.742)	(3.360)	(0.816)
ONGC	0.0012	0.0562	0.000 5	-0.0006	0.00015	0.310	0.385	-0.0026	0.0016
	(2.146)	(1.770)	(0.970)	(-0.744)	(8.896)	(8.547)	(9.345)	(-0.741)	(0.432)
WIPRO	-0.0030	0.0041	0.0172*	-0.2749*	0.00158	0.142	0.525	0.8097*	1.0095*
	(-0.842)	(1.113)	(4.572)	(-3.807)	(6.561)	(8.411)	(18.273)	(4.733)	(6.276)
ICICI	0.0002	0.0258	0.2482*	0.0008	0.0002	0.248	0.587	0.0824*	0.0002
	(4.397)	(1.485)	(3.265)	(0.789)	(5.741)	(9.265)	(36.541)	(3.852)	(0.781)

Note: ExpD and NextD indicate volatility effect and price reversal effect. Figures in parentheses are t values. * indicates 1 % level of significance.

Table 8: Empirical Results of Futures Expiration Day Volatility Effect with GARCH (1, 1) Model: Student - t Distribution

Mean Equation					Variance Equation				
Name of the stock	Constant	L.return	ExpD	NextD	Constant	ARCH(1)	GARCH(1	ExpD	NextD
ACC	0.00017	0.0173	0.0048*	0.0002	0.00147	0.150	0.830	-0.984*	0.5365
	(3.421)	(0.694)	(5.920)	(0.073)	(2.423)	(5.679)	(32.462)	(-5.172)	(3.276)
BHEL	0.00125	0.0024	0.1036*	0.0229	0.00066	0.203	0.690	0.0036*	0.0006

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	(2.596)	(2.376)	(3.257)	(0.836)	(4.098)	(5.394)	(14.960)	(2.096)	(0.123)
BPCL	-0.00237	0.0183	0.5018*	-0.0079*	0.00030	0.084	0.865	0.0059*	0.0063*
	(-0.409)	(0.754)	(5.640)	(-2.876)	(2.353)	(4.130)	(27.130)	(2.058)	(3.237)
CIPLA	0.00188	0.0379	0.0005	0.0052	0.00145	0.277	0.457	0.0085	-0.6642
	(0.674)	(1.940)	(0.723)	(0.922)	(5.555)	(4.212)	(7.060)	(1.123)	(-0.587)
DRREDDY	0.00035	0.0324	0.0015*	0.0103*	-0.00032	0.369	0.521	0.1087*	0.0911
	(0.882)	(1.221)	(3.773)	(2.062)	(-9.164)	(4.500)	(16.538)	(3.581)	0.521)
GRASIM	0.00085	0.0400	-0.0003	0.0035	0.00783	0.132	0.844	0.1027*	0.0001
	(2.030)	(1.550)	(-0.159)	(1.702)	(1.698)	(5.611)	(33.099)	(2.097)	(0.587)
HDFC	0.00024	0.0073	0.0078	0.0002	5.52E-05	0.226	0.619	0.0042	0.0082
	(0.914)	(0.575)	(0.681)	(1.025)	(4.681)	(5.003)	(11.306)	(0.052)	(0.023)
INFOSYS	0.00054	-0.0064	0.8922*	0.0048*	0.00062	0.089	0.892	0.6587*	-4.002*
	(0.684)	(-0208)	(0.584)	(2.809)	(19.083)	(3.996)	(25.142)	(5.423)	(-3.045)
ITC	0.00029	0.0118	0.1045*	0.0079*	0.00031	0.129	0.724	0.982*	-2.756*
	(0.654)	(0.477)	(4.046)	(3.052)	(16.213)	(5.764)	(41.506)	(3.456)	(-5.218)
M&M	0.00011	0.0942	0.0002	0.0004	0.00059	0.148	0.757	0.0003	0.0001
	(2.101)	(3.584)	(0.298)	(1.254)	(4.097)	(4.828)	(18.880)	(0.875)	(1.219)

Note: ExpD and NextD indicate volatility effect and price reversal effect. Figures in parentheses are t values. * indicates 1 % level of significance.

Table 8 (contd.,)

Mean Equation					Variance Equation				
Name of the stock	Constan	L.return	ExpD	NextD	Constant	ARCH(1)	GARCH(1	ExpD	NextD
RANBAXY	-0.00151	0.0176	0.0079*	0.0024	0.00012	0.207	0.530	0.0251*	0.0058
	(-3.366)	(0.650)	(2.561)	(1.233)	(5.055)	(4.099)	(7.660)	(6.0683)	(0.399)
RALIANCE	0.00156	0.0176	0.5009*	0.0006	0.00083	0.190	0.622	4.9361*	0.0058
	(3.366)	(0.165)	(9.078)	(0.187)	(4.396)	(5.155)	(9.693)	(8.701)	(0.168)
SATYAM	0.00038	0.0200	0.0005	0.0501*	0.00951*	0.163	0.719	2.4161*	-0.425*
	(0.639)	(0.856)	(1.957)	(2.552)	(4.403)	(4.740)	(15.703)	(6.411)	(-7.333)
SBI	0.00143	0.0105	0.0053*	0.0689*	1.05E-05	0.098	0.876	0.9321*	-0.255*
	(3.009)	(0.421)	(4.068)	(3.112)	(2.060)	(5.339)	(41.741)	(3.144)	(-0.879)
TATAPOWER	0.00129	0.0882	-0.048*	0.0066*	2.27E-05	0.156	0.806	2.0012*	0.0008

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	(2.685)	(3.475)	(-7.846)	(2.800)	(3.019)	(5.849)	(28.401)	(4.141)	(0.466)
HCLTECH	0.00340	0.0322	0.0053*	0.0059	3.58E-05	0.126	0.830	0.8791*	0.0008
	(0.567)	(1.146)	(5.630)	(0.876)	(2.929)	(4.520)	(27.813)	(5.429)	(0.178)
HEROHONDA	0.00042	0.0897	0.0082*	0.0013	0.00057	0.168	0.725	0.5032*	0.0007
	(0.768)	(0.968)	(2.597)	(0.575)	(3.563)	(4.547)	(13.970)	(3.097)	(0.162)
NATIONALU	0.00014	0.0211	0.0660*	-0.0022	5.75E-05	0.137	0.852	0.0061*	0.0008
	(2.366)	(0.757)	(2.512)	(-0.930)	(4.501)	(5.246)	(35.149)	(2.258)	(0.242)
ONGC	0.00914	0.0509	0.0036	-0.0030	2.37E-05	0.150	0.813	0.0015	0.0078
	(1.833)	(1.750)	(0.655)	(-1.229)	(2.89)	(4.839)	(26.807)	(0.223)	(0.234)
WIPRO	0.00029	0.0247	0.0141*	-0.0802	0.00027	0.234	0.360	2.5612*	0.007*
	(5.265)	(0.959)	(3.647)	(-6.178)	(5.694)	(3.675)	(4.388)	(8.428)	(2.894)
ICICI	0.00079	0.0002	0.0031	0.0214	0.00783	0.182	0.738	0.2087*	0.0001
	(2.030)	(1.250)	(0.159)	(1.702)	(3.749)	(4.613)	(33.099)	(3.097)	(0.587)

Note: ExpD and NextD indicate volatility effect and price reversal effect. The figures in parentheses are t values. * indicates 1 % level of significance

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